2.7 Classwork: Systems of Equations and Market Equilibrium (no calculator) — Markscheme

1. [Maximum mark: 13]

   Demand D: $p = 50 - 3q$. Supply S: $p = 10 + 2q$. Quantity q in thousands of cases per month, price p in euros.

   (a) $q \approx 8$ (thousand cases) **A1**

   $p \approx 26$ (euros) **A1**

   Accept $7 \leq q \leq 9$ *and* $24 \leq p \leq 28$. *Accept the point written as coordinates,* $(8,\ 26)$.

   (b) $3q + p = 50$ and $2q - p = -10$ **A1**

   Accept any equivalent pair, for example $-2q + p = 10$. *Both equations must be correct for the A1.*

   Adding the equations: $5q = 40$ **M1**

   $q = 8$ (thousand cases per month) **A1**

   $p = 50 - 3(8) = 26$ (euros) **A1**

   M1 for adding or subtracting the equations (after scaling, if used) so that one variable is eliminated. Accept eliminating q first: $2(3q+p) - 3(2q-p)$ *gives* $5p = 130$. *Substitution in place of elimination earns M0, then A1A1 for correct values. A correct answer with no working earns A1 for each value only (M0A1A1). FT the final A1 from the candidate's q.*

   (c) gradient $= 2$ **A1**

   For each extra 1000 cases per month supplied, the price rises by 2 euros. **R1**

   Accept the reverse reading: for each 2 *euro rise in price, suppliers offer* 1000 *more cases. The R1 needs both the direction (increase) and the units (euros per thousand cases). "The line goes up by 2" with no context earns R0.*

   (d) Substituting $p = 20$ into both equations **M1**

   Demand: $20 = 50 - 3q$, so $q = 10$ (thousand cases). Supply: $20 = 10 + 2q$, so $q = 5$ (thousand cases). **A1**

   Shortage $= 10 - 5 = 5$ thousand cases per month **A1**

   The A1 for quantities needs both values. Accept 5000 *cases. A negative answer,* -5, *or "a surplus of 5", earns the final A0. FT the shortage from the candidate's two quantities, provided demand exceeds supply.*

   (e) Triangle with vertices $(0,\ 50)$, $(0,\ 26)$, $(8,\ 26)$: base 8, height $50 - 26 = 24$ **M1**

   Area $= \frac{1}{2} \times 8 \times 24 = 96$, so the consumer surplus is 96 000 euros **A1**

M1 for the correct base and height (seen in an area calculation), or for the correct triangle shaded on the graph. Accept 96 *thousand euros.* 96 *euros (units ignored) earns A0. FT from the candidate's equilibrium point in (b).*

Wrong methods: the triangle down to the q-axis, $\frac{1}{2} \times 8 \times 26 = 104$*, earns M0A0. The rectangle* $8 \times 24 = 192$ *earns M0A0. The producer surplus,* $\frac{1}{2} \times 8 \times 16 = 64$*, earns M0A0.*

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2. [Maximum mark: 11]

(a) $x + y = 10\,000$ **A1**

$0.03x + 0.07y = 540$ **A1**

Accept $3x + 7y = 54\,000$ *or* $\frac{3}{100}x + \frac{7}{100}y = 540$*.* $3x + 7y = 540$ *(percentages not converted) earns A0.*

(b) $3x + 7y = 54\,000$ and $3x + 3y = 30\,000$ **M1**

Subtracting: $4y = 24\,000$, so $y = 6000$ **A1**

$x = 10\,000 - 6000 = 4000$ **A1**

Giulia puts 4000 euros in the bond and 6000 euros in the fund.

M1 for scaling one or both equations so that a variable can be eliminated, and adding or subtracting. Substitution earns M0, then A1A1 for correct values. FT from the candidate's equations in (a). A correct answer with no working earns A1A1 only.

Wrong method: swapping the rates ($0.07x + 0.03y = 540$*) gives* $x = 6000$*,* $y = 4000$*; FT from (a) applies only if the swap appears there, otherwise M1A0A0.*

Extension

(c)(i)
$$\begin{aligned} x + y + z &= 20\,000 \\ 0.02x + 0.04y + 0.06z &= 740 \\ y - z &= 2000 \end{aligned}$$
A1A1

A1 for the interest equation (accept $x + 2y + 3z = 37\,000$ *or* $2x + 4y + 6z = 74\,000$*). A1 for both the total and the difference equations (*$y = z + 2000$ *is fine).*

(c)(ii) Scale the interest equation: $x + 2y + 3z = 37\,000$.

Subtract the total equation: $y + 2z = 17\,000$ **M1**

Subtract $y - z = 2000$: $3z = 15\,000$, so $z = 5000$ **A1**

$y = 5000 + 2000 = 7000$ **A1**

$x = 20\,000 - 7000 - 5000 = 8000$ **A1**

M1 for eliminating one variable to reach a correct equation in the other two. Any order of elimination is fine. FT from the candidate's equations in (c)(i). Check: $160 + 280 + 300 = 740$.